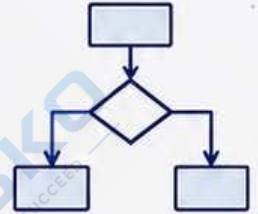


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# Question Bank

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# Software Engineering

## Unit I - Introduction to Software Engineering and Software Development Models

- **Answers in one or two lines (1 mark questions)**

1. Define Software.
2. Define Software Engineering.
3. What is a software product?
4. State any two characteristics of software.
5. Name any two types of software.
6. What is system software?
7. What is application software?
8. What is the need for Software Engineering?
9. Expand SDLC.
10. Define Software Development Life Cycle.
11. Name the phases of SDLC.
12. What is a software process model?
13. Define Waterfall Model.
14. Define Prototype Model.
15. Define Spiral Model.
16. Which model follows a sequential development approach?
17. What is a prototype?
18. What is risk analysis in the Spiral Model?
19. Name any two Software Development Models.
20. What is the purpose of SDLC?

● **Short Answer Questions (3 marks questions)**

1. Define Software Engineering and explain its need.
2. Explain the characteristics of software.
3. Explain different types of software.
4. Explain the importance of Software Engineering.
5. Explain Software Development Life Cycle (SDLC).
6. Explain the phases of SDLC.
7. Explain the concept of Software Process Models.
8. Explain the Waterfall Model with its phases.
9. Explain the Prototype Model.
10. Explain the Spiral Model.
11. Differentiate between Waterfall Model and Prototype Model.
12. Differentiate between Waterfall Model and Spiral Model.
13. Explain the advantages of the Waterfall Model.
14. Explain the advantages and disadvantages of the Prototype Model.
15. Explain the advantages and disadvantages of the Spiral Model.

● **Long Answer Questions (6 marks questions)**

1. Define Software Engineering and explain its characteristics, need, and importance.
2. Explain different types of software with suitable examples.
3. Explain SDLC and all its phases in detail with a suitable diagram.
4. Explain Software Development Process Models and their importance.
5. Explain the Waterfall Model with a neat diagram, advantages, and disadvantages.
6. Explain the Prototype Model with a neat diagram, advantages, and disadvantages.
7. Explain the Spiral Model with a neat diagram, advantages, and disadvantages.
8. Compare Waterfall, Prototype, and Spiral Models.
9. Explain the different phases of SDLC and their activities.
10. Explain the Waterfall, Prototype, and Spiral Models and discuss their suitable applications.

## Unit II - Software Requirements and Requirements Engineering

- **Answers in one or two lines (1 mark questions)**

1. Define Software Requirement.
2. What is Requirements Engineering?
3. What is a Functional Requirement?
4. What is a Non-functional Requirement?
5. Define User Requirement.
6. Define System Requirement.
7. What is Requirements Elicitation?
8. What is Requirements Analysis?
9. What is Requirements Validation?
10. What is Requirements Management?
11. What is a Software Requirements Specification (SRS)?
12. What is the purpose of requirements engineering?
13. Give one example of a functional requirement.
14. Give one example of a non-functional requirement.
15. What is a stakeholder?
16. Who is a software user?
17. What is requirements documentation?
18. Why is requirements validation required?
19. What is requirements change?
20. Name the major activities of the requirements engineering process.

● **Short Answer Questions (3 marks questions)**

1. Explain Software Requirements.
2. Explain Functional Requirements with examples.
3. Explain Non-functional Requirements with examples.
4. Differentiate between Functional and Non-functional Requirements.
5. Explain User Requirements.
6. Explain System Requirements.
7. Differentiate between User Requirements and System Requirements.
8. Explain Requirements Engineering and its importance.
9. Explain the Requirements Engineering Process.
10. Explain Requirements Elicitation and Analysis.
11. Explain Requirements Validation.
12. Explain Requirements Management.
13. Explain the role of stakeholders in requirements engineering.
14. Explain the importance of requirements documentation.
15. Explain problems associated with changing software requirements.

● **Long Answer Questions (6 marks questions)**

1. Explain Software Requirements and their different types in detail.
2. Explain Functional and Non-functional Requirements with suitable examples.
3. Differentiate between Functional and Non-functional Requirements with examples.
4. Explain User Requirements and System Requirements in detail.
5. Explain the Requirements Engineering Process with a suitable diagram.
6. Explain Requirements Elicitation and Analysis in detail.
7. Explain Requirements Validation, its objectives, and techniques.
8. Explain Requirements Management and its importance.
9. Explain the complete process of Requirements Engineering from elicitation to management.
10. Explain the role of requirements engineering in successful software development.

### Unit III - Risk Management and Design Engineering

- **Answers in one or two lines (1 mark questions)**

1. Define software risk.
2. What is risk management?
3. What is risk identification?
4. What is a risk strategy?
5. What is an RMMM Plan?
6. Expand RMMM.
7. What is risk monitoring?
8. What is risk mitigation?
9. Define Design Engineering.
10. What is software design?
11. What is the design process?
12. What is design quality?
13. Define software architecture.
14. What is data design?
15. What is an architectural pattern?
16. What is User Interface Design?
17. What are Golden Rules of UI Design?
18. What is a system model?
19. What is UML?
20. Expand UML.
21. What is a UML diagram?
22. Name any two UML diagrams.
23. What is a Use Case Diagram?
24. What is a Class Diagram?
25. What is risk assessment?

● **Short Answer Questions (3 marks questions)**

1. Explain software risk.
2. Explain the importance of risk management.
3. Explain different software risks.
4. Explain risk identification.
5. Explain risk strategies.
6. Explain the RMMM Plan.
7. Explain the components of an RMMM Plan.
8. Explain the software design process.
9. Explain design quality.
10. Explain Software Architecture.
11. Explain the importance of software architecture.
12. Explain data design.
13. Explain architectural patterns.
14. Explain User Interface Design.
15. Explain the Golden Rules of User Interface Design.
16. Explain the steps involved in interface design.
17. Explain system models.
18. Explain UML and its importance.
19. Explain any three UML diagrams.
20. Differentiate between software architecture and software design.

● **Long Answer Questions (6 marks questions)**

1. Explain Risk Management and its importance in software engineering.
2. Explain different types of software risks and risk identification techniques.
3. Explain risk strategies and the RMMM Plan in detail.
4. Explain the Risk Mitigation, Monitoring, and Management (RMMM) Plan with a suitable example.
5. Explain the Software Design Process and characteristics of a good design.
6. Explain Design Quality and the factors affecting software design quality.
7. Explain Software Architecture, Data Design, and Architectural Patterns.
8. Explain User Interface Design, its Golden Rules, and interface design steps.

9. Explain System Models and UML Diagrams with suitable examples.
10. Explain the complete design engineering process from software design to user interface design.

#### **Unit IV - Inheritance and Interfaces**

● **Answers in one or two lines (1 mark questions)**

1. Define Software Testing.
2. What is a Testing Strategy?
3. What is the purpose of Software Testing?
4. What is Black Box Testing?
5. What is White Box Testing?
6. What is Validation Testing?
7. What is System Testing?
8. What is Debugging?
9. What is a test case?
10. What is a test plan?
11. What is the difference between testing and debugging?
12. What is conventional software testing?
13. What is the objective of validation testing?
14. What is the objective of system testing?
15. Name any two types of software testing.
16. What is a testing strategy?
17. What is a software error?
18. What is a software defect?
19. What is a failure in software testing?
20. What is the role of debugging in software development?

● **Short Answer Questions (3 marks questions)**

1. Explain Software Testing.
2. Explain the importance of Software Testing.
3. Explain a strategic approach to software testing.
4. Explain the major activities involved in a testing strategy.
5. Explain test strategies for conventional software.
6. Explain Black Box Testing with an example.
7. Explain White Box Testing with an example.
8. Differentiate between Black Box Testing and White Box Testing.
9. Explain Validation Testing.
10. Explain the objectives of Validation Testing.
11. Explain System Testing.
12. Explain the objectives of System Testing.
13. Differentiate between Validation Testing and System Testing.
14. Explain Debugging.
15. Explain the difference between Software Testing and Debugging.
16. Explain the characteristics of an effective testing strategy.
17. Explain the role of test cases in software testing.
18. Explain the importance of systematic testing.
19. Explain the process of identifying errors during testing.
20. Explain the relationship between testing and debugging.

● **Long Answer Questions (6 marks questions)**

1. Explain Software Testing and the strategic approach to software testing in detail.
2. Explain the complete testing strategy for conventional software.
3. Explain Black Box Testing and White Box Testing with suitable examples.
4. Differentiate between Black Box Testing and White Box Testing in detail.
5. Explain Validation Testing, its objectives, and process.
6. Explain System Testing with suitable examples.
7. Differentiate between Validation Testing and System Testing.
8. Explain the Art of Debugging and the debugging process in detail.
9. Explain the relationship between Software Testing and Debugging.

10. Explain the complete software testing process from test planning to debugging.